

# MANAS SHUKLA

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## EXPERIENCE

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*Senior Systems Engineer, Sensata Technologies* | Attleboro, MA, USA

**Jan 2022 – Mar 2025**

- Led software development for multimillion-dollar, multi-station complex machinery performing operations such as calibration, characterization, pick and place, etc. by designing control logic and operational sequences in compliance with the URS.
- Developed control systems using various PLC platforms, enhancing automation processes and improving equipment cycle time by roughly 15% while reducing operational errors by ~20% through a modular architecture design philosophy.
- Designed HMI interfaces using WinForms and TwinCAT; built SCADA dashboards for intuitive equipment control & monitoring.
- Created and optimized SQL database queries, including table creation and data management, leveraged Python scripting with SQL to interface equipment with the MES, enhancing data-driven decision-making, equipment traceability, and performance tracking.
- Configured servo drives, actuators, and VFDs (Schneider, Beckhoff, Siemens, etc.) and implemented safety systems for robots, gantries, pneumatics, and conveyors, ensuring compliance with industry standards using protocols like EtherCAT, and Ethernet/IP.
- Managed project schedules, costs, scope, and objectives while coordinating with suppliers and cross-functional team members across the globe to ensure seamless automation integration and scalability from conceptual development through installation.
- Conducted statistical analyses (GRR and ANOVA) to validate manufacturing process improvements, ensuring equipment accuracy.

*Machine Learning Engineer, Universal Logic Inc.* | Charlotte, NC

**May 2018 – Jan 2022**

- Developed and deployed various custom Machine learning and Deep learning algorithms and reactive robot control systems for object detection, motion planning, and industrial automation tasks.
- Collected, processed, and analyzed data from diverse online and in-house sources using clustering methods, dimensionality reduction, and Python-based pipelines for robust model training.
- Developed scalable data engineering solutions using Python, C++, and AI/ML libraries such as TensorFlow, PyTorch, and Keras on AWS infrastructure, including S3 for data management and EC2 GPU instances for high-performance inference.
- Designed and maintained centralized databases and web-based interfaces for efficient data storage, access, and visualization.
- Determined optimal clustering methods and intelligence training strategies to improve robot decision-making and task execution.
- Conducted continual research on emerging ML techniques and robotics trends to refine model accuracy and system integration.

*Research Assistant, Real-Time & Embedded Systems Lab (mLab)* | Philadelphia, PA

**Jun 2017 – May 2018**

- Developed an end-to-end toolchain to streamline the process from mission planning to deployment for drone operations.
- Designed and implemented control algorithms to generate waypoints for user-defined trajectories, including multi-drone coordination, dynamic obstacle avoidance, and other advanced flight maneuvers.
- Built a simulation environment using CAD modeling integrated with ROS/Gazebo for testing and validating flight scenarios.

*CAD Application Developer, Quest-Global Inc.* | Bangalore, India and Bristol, United Kingdom

**Jun 2011 – Oct 2013**

- Collaborated with a 25-member team to deploy a multidisciplinary cost-optimization framework for turbine blade components used in Rolls-Royce aero-engines.
- Gathered and analyzed requirements from designers, engineers, and manufacturers to develop 40+ parametric features and an intuitive Graphical User Interface (GUI) using NX design automation tools.
- Represented the Quality and Improvements team in the UK for five months, leading project timelines and co-developing geometry workflows using Simulia Isight, achieving a 100% Customer Satisfaction (C-Sat) score.
- Spearheaded two new automation projects ahead of schedule, authored 10+ technical articles for the company's Knowledge Portal (viewed over 500 times), and trained four new employees, earning Quest's prestigious Performance Recognition Award.

## EDUCATION

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*Master of Science, Mechanical Engineering, University of Pennsylvania*

**Aug 2014 - May 2016**

*Bachelor of Technology, Mechanical Engineering, Vellore Institute of Technology University*

**Jul 2007 - Jun 2011**

## SKILLS

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**Expertise:** Industrial Automation | Control Systems | Robotics | Machine Learning | Deep Learning | Project Management

**Programming Languages:** Python | R | C, C++ | C# | IEC 61131-3 (Ladder Logic, Structured Text) | LabVIEW | MATLAB

**Control Systems:** Beckhoff | TwinCAT | Siemens | TIA Portal | Mitsubishi | MELSEC GXWorks